

PENSE-BÊTE : Dashboard MQTT Local & Service Systemd

1. Structure du projet (Dossier ~/natacha-dashboard)

Le dossier contient trois éléments indispensables :

- `index.html` : La page web du dashboard de monitoring.
- `paho-mqtt.js` : La bibliothèque JavaScript locale pour la gestion du protocole MQTT via WebSockets.
- `favicon.ico` : L'icône du site (optionnelle).

2. Code Source de `index.html` (Version Définitive)

HTML

```
<!DOCTYPE html>
<html lang="fr">
<head>
  <meta charset="UTF-8">
  <title>Natacha - MQTT Monitor</title>
  <script src="paho-mqtt.js"></script>
  <style>
    body { font-family: sans-serif; background: #121212; color: #e0e0e0;
margin: 0; padding: 20px; }
    h1 { color: #4CAF50; }
    #status { font-weight: bold; margin-bottom: 15px; }
    .connected { color: #4CAF50; }
    .disconnected { color: #F44336; }
    #messages { background: #1e1e1e; border: 1px solid #333; padding: 15px;
height: 400px; overflow-y: scroll; font-family: monospace; border-radius: 5px; }
    .msg { margin-bottom: 8px; border-bottom: 1px solid #2a2a2a; padding-
bottom: 4px; }
    .topic { color: #64B5F6; font-weight: bold; }
    .payload { color: #81C784; }
  </style>
</head>
<body>

  <h1>🌐 Natacha - Dashboard MQTT Local</h1>
  <div id="status" class="disconnected">État : Connexion en cours...</div>

  <h3>Flux des messages en direct (natacha/#) :</h3>
  <div id="messages"></div>

  <script>
    window.onload = function() {
      const brokerHost = window.location.hostname || "192.168.1.80";
      const brokerPort = 9001;
      const clientId = "natacha_web_monitor_" +
Math.random().toString(16).substring(2, 8);

      console.log(`Connexion à ws://${brokerHost}:${brokerPort} ...`);
```

```

const client = new Paho.Client(brokerHost, Number(brokerPort),
clientId);

client.onConnectionLost = function (responseObject) {
    document.getElementById("status").className = "disconnected";
    document.getElementById("status").innerText = "État : Déconnecté
(" + responseObject.errorMessage + ")";
    setTimeout(connectMQTT, 4000);
};

client.onMessageArrived = function (message) {
    let container = document.getElementById("messages");
    let time = new Date().toLocaleTimeString();
    let html = `<div class="msg">[${time}] <span class="topic">${
message.destinationName}</span> : <span class="payload">${
{escapeHtml(message.payloadString)}</span></div>`;
    container.innerHTML += html;
    container.scrollTop = container.scrollHeight;
};

function connectMQTT() {
    client.connect({
        onSuccess: function () {
            document.getElementById("status").className =
"connected";
            document.getElementById("status").innerText = "État :
Connecté au broker Mosquitto (" + brokerHost + ":" + brokerPort + ")";
            client.subscribe("natacha/#");
        },
        onFailure: function (message) {
            document.getElementById("status").className =
"disconnected";
            document.getElementById("status").innerText = "État :
Échec de connexion (" + message.errorMessage + ")";
            setTimeout(connectMQTT, 4000);
        },
        useSSL: false
    });
}

connectMQTT();

function escapeHtml(text) {
    return text.replace(/&/g, "&").replace(/</g,
"&lt;").replace(/>/g, "&gt;");
}
</script>
</body>
</html>

```

3. Configuration du Broker Mosquitto (Docker)

Fichier mosquitto.conf :

Plaintext

```

persistence true
persistence_location /mosquitto/data/
log_dest stdout

listener 1883

```

```
allow_anonymous true
```

```
listener 9001  
protocol websockets
```

Lancement du conteneur Docker avec les ports mappés (1883 et 9001) :

Bash

```
docker run -d \  
  --name mosquitto \  
  -p 1883:1883 \  
  -p 9001:9001 \  
  -v ~/mqtt/config/mosquitto.conf:/mosquitto/config/mosquitto.conf \  
  -v ~/mqtt/data:/mosquitto/data \  
  eclipse-mosquitto
```

4. Persistance Systemd (Service Web Python)

1. Créer le fichier de service :

Bash

```
sudo nano /etc/systemd/system/natacha-dashboard.service
```

2. Y placer le contenu suivant :

Ini, TOML

```
[Unit]  
Description=Natacha Local MQTT Dashboard  
After=network.target
```

```
[Service]  
Type=simple  
User=vieil  
WorkingDirectory=/home/vieil/natacha-dashboard  
ExecStart=/usr/bin/python3 -m http.server 8090  
Restart=always
```

```
[Install]  
WantedBy=multi-user.target
```

3. Activer et démarrer le service :

Bash

```
sudo systemctl daemon-reload  
sudo systemctl enable natacha-dashboard.service  
sudo systemctl start natacha-dashboard.service
```

4. Commandes utiles de gestion :

- Vérifier l'état : `sudo systemctl status natacha-dashboard.service`
- Redémarrer : `sudo systemctl restart natacha-dashboard.service`
- Arrêter : `sudo systemctl stop natacha-dashboard.service`

5. Commandes de Diagnostic Rapide

- Tester l'ouverture du port WebSocket (9001) :

Bash

```
nc -zv 192.168.1.80 9001
```

- Publier un message test en MQTT :

Bash

```
mosquitto_pub -h 192.168.1.80 -p 1883 -t "natacha/reponse" -m "Test direct"
```

- Accès au Dashboard dans le navigateur : [<http://192.168.1.80:8090>]
(<http://192.168.1.80:8090>)